Submission Summary

Conference Name

4th International Conference on Knowledge Engineering and Communication Systems (ICKECS2026)

Track Name

Knowledge Engineering

Paper ID

1086

Paper Title

Sentinel: A Human-in-the-loop AI Framework for Electoral De-Duplication and Voter Identity Verification

Abstract

Ensuring the precision of elections is crucial for maintaining respect and trust in democracy and other governmental procedures [12]. Duplicate voter representations, arising from numerous sources like administrative problems and identity fraud, present a significant challenge in large scale democracies. But facial recognition systems present in the contemporary world may not solve all the issues of de-duplication, new issues can develop from it like removal of real and genuine voters. This research paper introduces Sentinel, an AI enabled technology and innovation [10] that uses the combination of deep learning as well a human in the loop mechanism. The system uses an SSD MobileNet V1 to generate 128-dimensional facial embeddings, which are then compared using the Euclidean distance to identify any duplicates in the database. Instead of automatically rejecting suspicious records they will be resolved by human intervention.

Created

1/14/2026, 2:48:38 PM

Last Modified

1/14/2026, 2:48:38 PM

Authors

Suhas Umesh (The National Institute of Engineering) <suhasu710@gmail.com>

Primary Subject Area

Artificial Intelligence, Expert Systems

Secondary Subject Areas

Computer Vision, Content Based Image Retrieval

Data Science and Predictive Analytics

Image Databases and Bio-informatics

Intelligent Information Processing Systems

NLP, Machine Learning and Soft Computing Applications

Submission Files

SSSZ Sentinel.docx (234.4 Kb, 1/14/2026, 2:45:26 PM)